

$$\begin{aligned}
& \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} \\
&= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{qq'} \delta_{q^2}^+(p_{12} - k) \delta_0^+(k)}{p_1^2 p_2^2 (k - p_2)^2 (k - p_1)^2}
\end{aligned}$$